SHOBBIKA.T
CSE - 3rd year

COMPUTER NETWORKS
OBSERVATION.

240701502.

S.NO	DATE	EXPERIMENT	T. SIGN
1.	23/7/26	Study various Networking commands used in Linux and Windows	
2.	23/7/26	Network Cables	
3.	24/8/26	Packet tracer - hub	
4.	24/8/26	LAN Configuration	
5.	11/9/26	Wire Shark	
6.	7/9/26	Hamming Code	
7.	7/9/26	Sliding Window	
8.	15/9/26	N-Map	
9.	22/9/26	Subnetting	
10.	22/9/26	Internetworking	
11.	28/9/26	Static Routing + RIP	
12.	28/9/26	TCP/UDP Sockets	
13.	5/10/26	Own Ping	
14.	5/10/26	RAW Socket Sniffing	

Completed 12/10/26

EX. NO: 1
DATE : 23/7/26

PRACTICAL - 1

Network Commands and Linux Commands

AIM: Study of Various Network commands used in Linux and Windows.

BASIC NETWORKING COMMANDS:

1. arp.a : Address resolution Protocol

Output:

Interface : 192.168.56.1 --- 0xb

Internet Address	physical Address	Type
192.168.56.255	ff-ff-ff-ff-ff-ff	Static
224.0.02	01-00-5e-00-00-02	Static

2. hostname: Displays the name of the computer

O/P: LAP-Shobbika

3. ip config /all : Info about TCP/IP connection

O/P: Windows ip configuration

Hostname . . . . . : LAP-Shobbika
Primary DNS suffix . . :
Node Type . . . . . : Mixed
Ip routing enabled . . . : No
WINS Proxy Enabled . . : No

4. nbtstat -a: Netbios over TCP/IP

O/P:

-a (adapterStatus) Lists the remote machine's name table given

-c (Cache) Lists the NBT's cache.

-n (names) Lists local NetBIOS names.

-RR (Release Refresh) Sends Name release packets to WINS and then, starts refresh.

5. netStat : (Network statistics)

O/P: Active Connections

Proto	Local Address	Foreign Address
TCP	127.0.0.1:44670	Lap-Shobbika:5140
TCP	127.0.0.1:44670	Lap-Shobbika:52097

6. nslookup: (name server Lookup)

O/P: Server: Unknown
Address: 172.16.8.1
Name: www.google.com
Address: 2001:4860:4826:770001:
2001:4860:4826:7700:1

7. pathping:

O/P: pathping [-g hostlist][-h maximum_hops]
[-i address][-n][-p period][-q num_queries]
[-w timeout][-4][-6] target_name.

8. ping: Packet Internet Grouper

O/P: Pinging LAPShabbike [fe80:db93:8480:2114:
+393%13] with
32 bytes of data,

9. route: IP routing table.

O/P: ROUTE: [-f][-p][-4][-6] command [destination]
[MASK network] [gateway] [METERIC metric]
[IF interface]

LINUX commands:

1. ip: Displays ip address.

O/P: Usage: ip [OPTIONS] object {COMMAND/help}
ip [-force] -batch filename

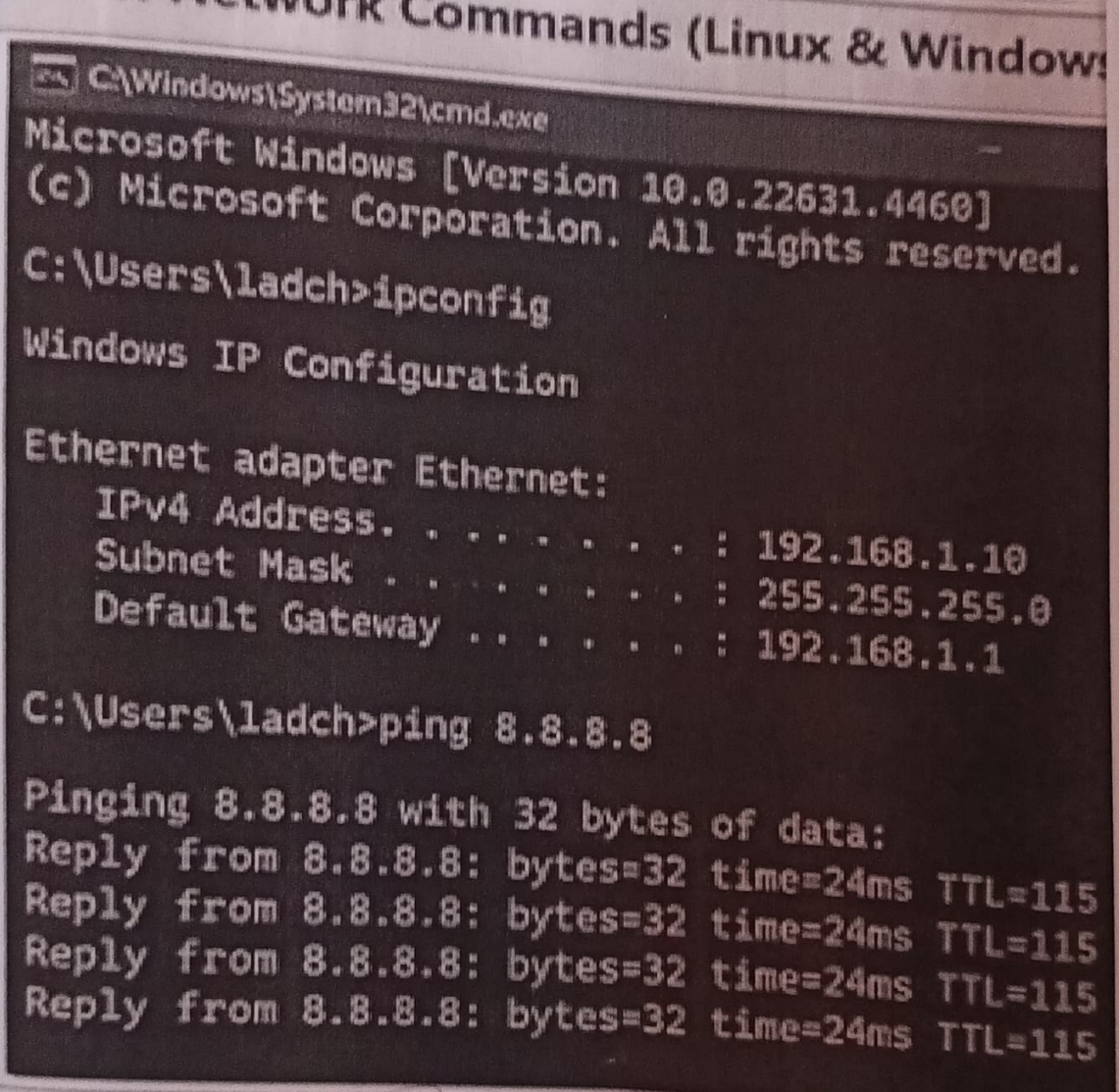
2. ip route

O/P: 10.0.2.2 dev enpr033 proto dhcp src
10.0.215 metric 100 10.0.2.0/24 dev
en P033 proto kernel scope link
src 10.0.2.15 metric 100

3. ifconfig: Configuring and troubleshooting networks

O/P: Windows IP Configuration
Unknown adapter McAfee VPN:
Media state . . . : Media disconnected
Connection-specific DNS Suffix

1. Network Commands (Linux & Windows



4. mtr: combines ping and traceroute commands

o/p:

host	Loss%	Snt	Last	Avg	best
1. 172.16.8.1	0.0%	134	0.1	0.2	0.1
2. 172.97.163.129	2.6%	134	3.9	5.2	1.1

5. tcpdump:

i> tcpdump -i eth0 -c 10 host 8.8.8.8

6. ping LAP.SHOBBIKA

o/p: pinging LAP-Shobbika [fe80::d093:8480:2114:4393:1339]

with 32 bytes of data

Reply from fe80::d093:8480:2114:4393%13: time<1ms

Result: The basic networking of Linux commands learned and executed successfully.

NETWORK CABLES

AIM: To study different types of network cables and to make an ethernet cross-over straight cable.

Types of network cables:

1. unshielded twisted pair cable
2. shielded twisted pair cable.
3. SSTP Cable
4. Coaxial cable
5. Fibre optic cable

Tools and ports required:

1. ethernet cable (CAT 5 / CAT 5e)
2. Crimping tool
3. Two RS 45 plugs
4. Plug shields (optional)

Procedure:

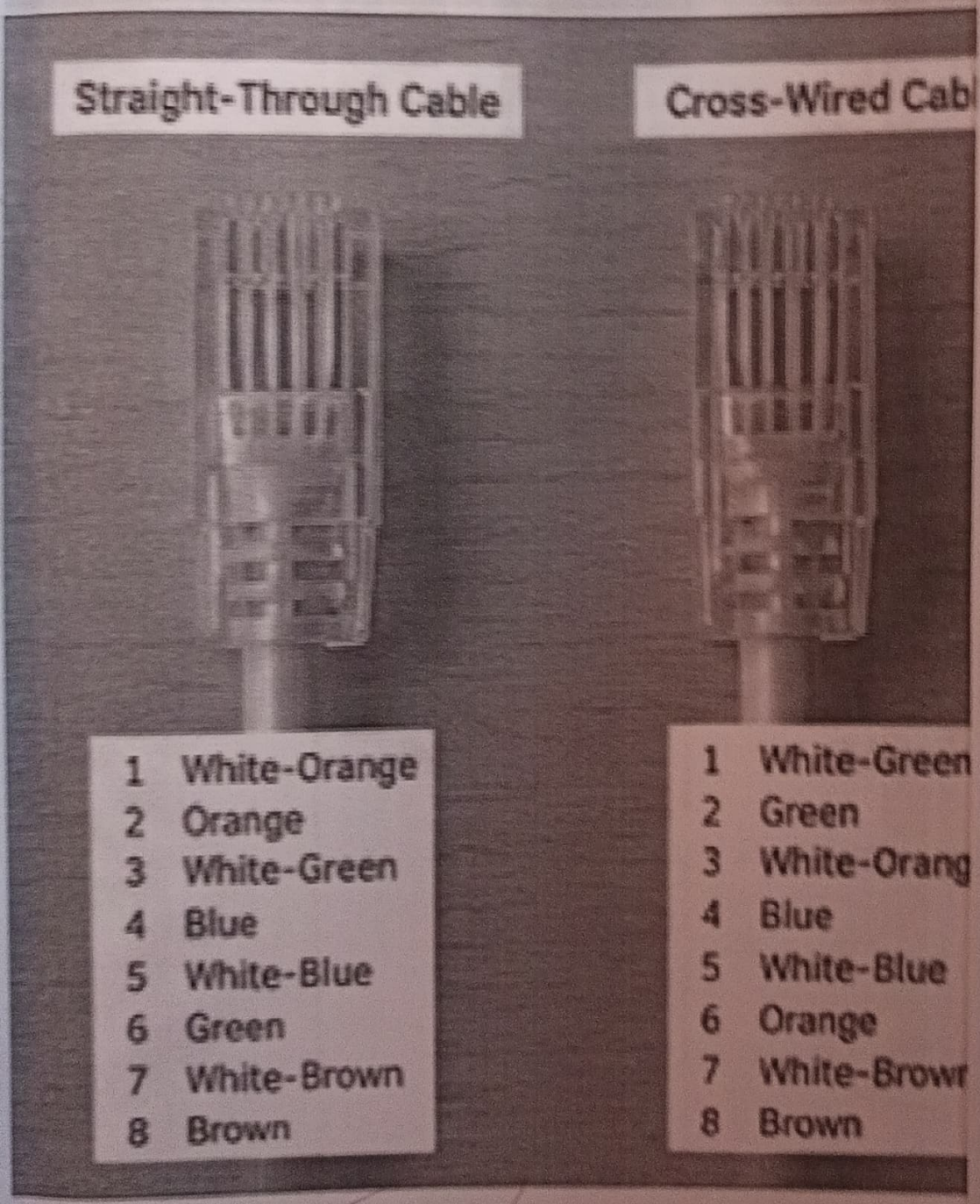
Study of network cables:

1. Identify different types of network cables.
2. Study their category, data, transmission rates, advantages, disadvantages and applications.
3. Observe the physical structure of each cable.

Making ethernet crossover / straight cables:

1. Thread the shields onto the cable
2. Strip approximately 1.5 cm of cable shielding from both ends using the crimping tools.
3. untangle the 4 twisted pairs of wires.
4. Arrange the wires according to the required wiring arrangements.
5. Align the wires evenly & insert into the plug.
6. Ensure that the end of the shielding reaches the notch of the RJ45 plug.
7. Insert the RJ45 plug into the crimping tool and press it carefully.

2. Network Cables



crimping tool and press it [illegible].

8. For a crossover cable, use arrangement A at one end and B at other end.

9. For a straight-through cable, use arrangement A at both ends.

10. Test the cable by connecting the required ethlines.

Result: Thus different types of Cables and Ethernet (cross-over / straight Cable was successfully studied.

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Ex: 3 Study of Cisco Packet tracer and network device Analysis

AIM: To study the installation and user interface of cisco packet traur and to understand its environment for designing a simple network devices such as hub and switch using cisco packet traur.

Requirements:

→ Cisco packet tracer
→ 4 Generic PCs
→ 1 hub
→ 1 Switch
→ Copper straight - through cables

Algorithm/Procedure:

1. Open Cisco packet tracer
2. Place 4 PCs and 1 Hub
3. Connect all PCs to the hub using Copper straight - through cable.
4. Configure the IP addresses.

PC	IP address	Subnet mask
PC0	192.168.1.1	255.255.255.0
PC1	192.168.1.2	255.255.255.0
PC2	192.168.1.3	255.255.255.0
PC3	1927.168.1.4	255.255.255.0

5. Select Add simple PDU
6. Click PC0 and the PC1.
7. Observe the packet flow.

Network using Switch:

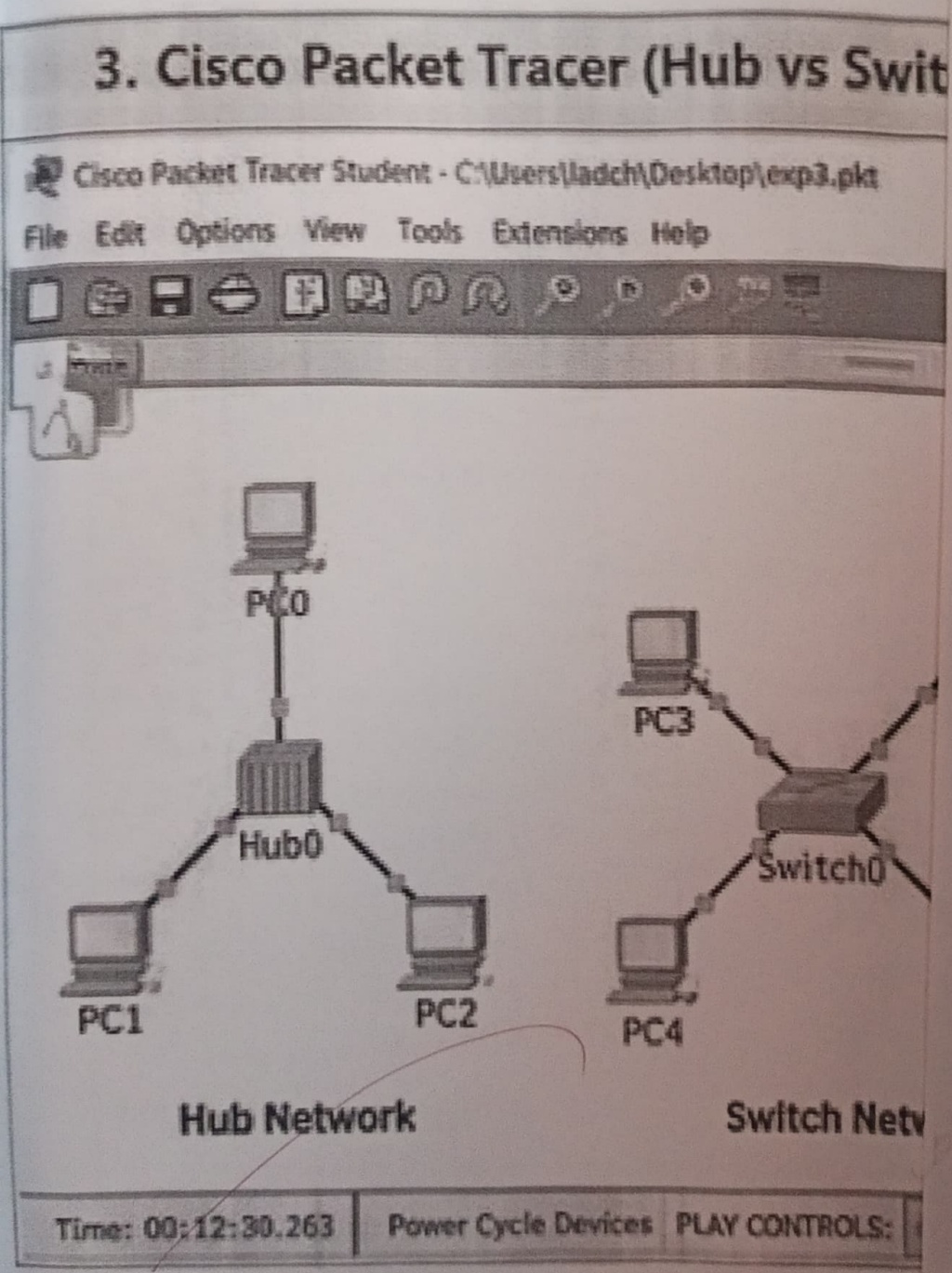
1. place 4 PCs and 1 switch, connect all through switch.
2. Assign IP address.
3. Send a PDU from PC0 to PC1.
4. Observe Packet flow.

Result: Thus Cisco packet traur was studied successfully and hub and switch connection was also studied successfully.

29/1/20

Ex:3 Study of Cisco Packet tracer and n...

3. Cisco Packet Tracer (Hub vs Swit



...ideas and studied necessary...

4/7/26

Ex4 LAN Setup using Switch

AIM: To setup and configure a LAN using a switch and ethernet Cables and Check connectivity between computers.

Requirments:

1. 4PCs
2. 1 Switch
3. 4 ethernet Cables.

Algorithm:

1. Connect 4PCs to a switch.
2. Assign IP addresses.
3. Check the connection using ping.
4. Share a folder between computers.
5. Check whether the shared folder can be accessed.

Procedure:

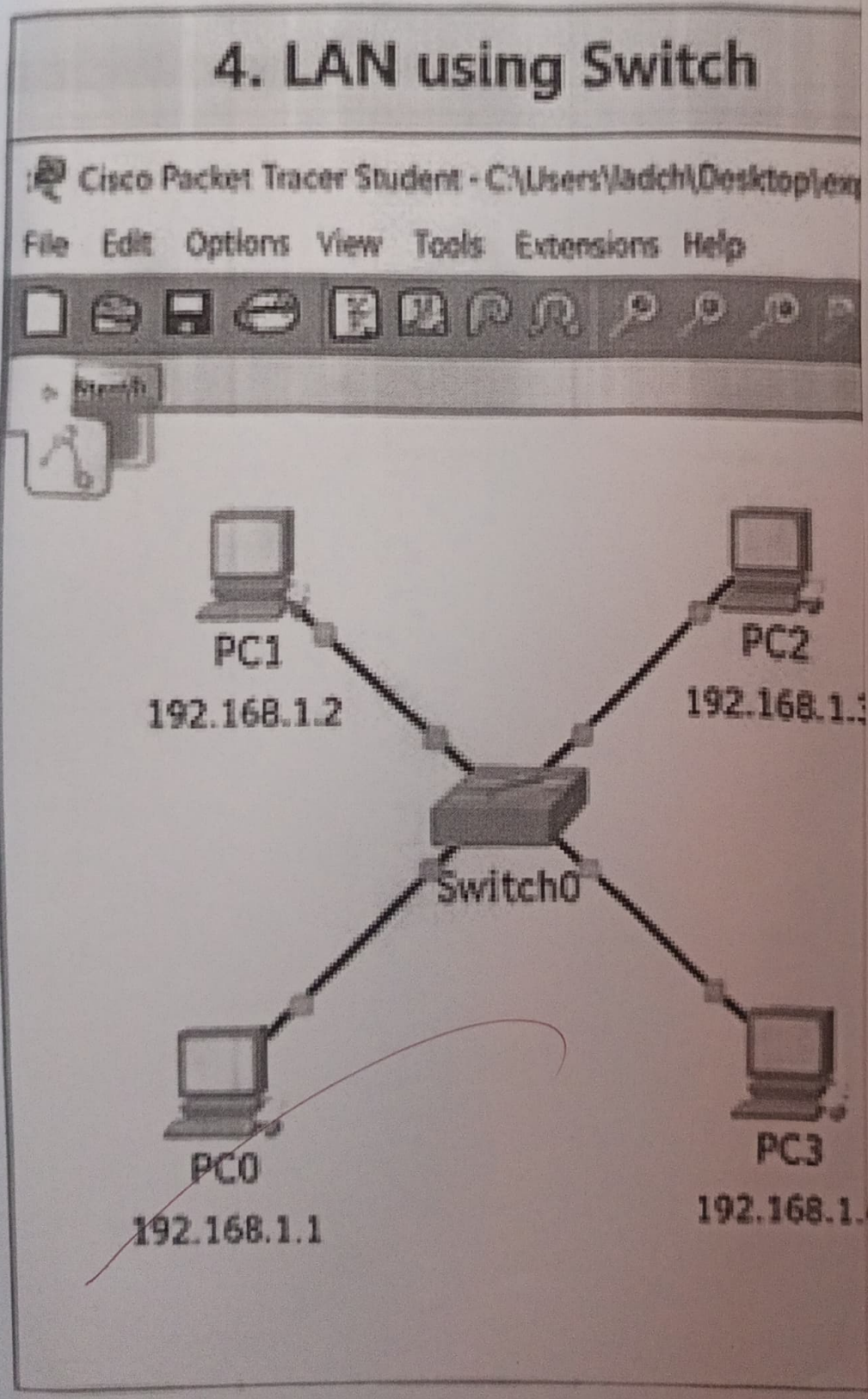
1. Connect 4 PCs to the switch using ethernet cables.
2. Assign the IP addresses.
3. Open cmd
4. check IP using:
   ipconfig
5. From PC1 check the connection to PC2 using Ping 10.1.1.2

Result: Thus, a LAN was successfully created and configured using a switch and ethernet Cables connectivity between the computers was verified using the ping command.

24/7/26

Ex4

4. LAN using Switch



command.

1/9/26

Ex:5

Wireshark

AIM: To study the wireshark packet capture tool and analyse packets at different layers of the protocol stack.

PROCEDURE:

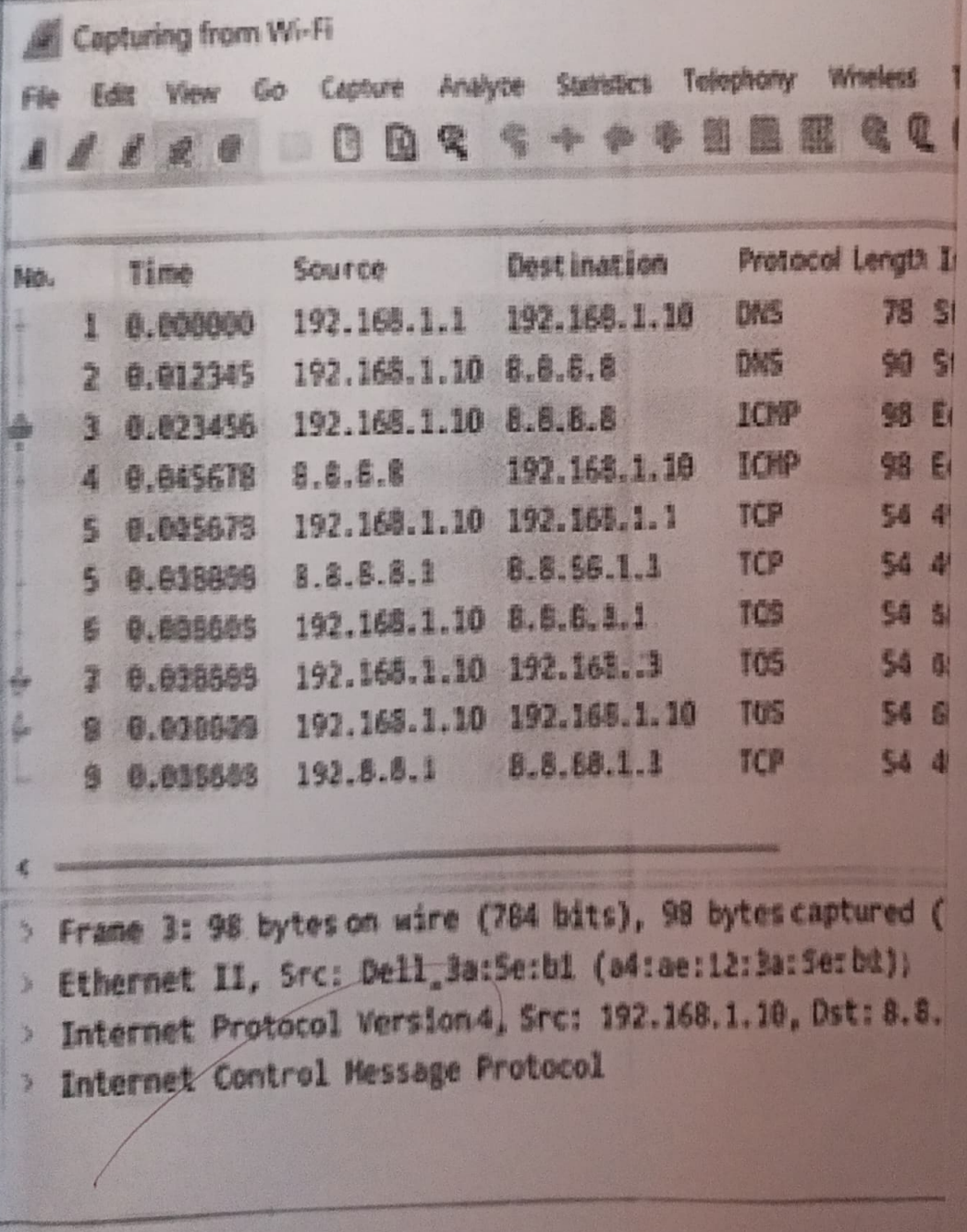
1. Open Wireshark
2. Select the local Area Connection / Ethernet Interface.
3. Start packet Capture.
4. Capture approximately 100 packets.
5. Stop the Capture.
6. Apply filters for:
   - → TCP / UDP
   - → ARP
   - → DNS
   - → HTTP
   - → IP / ICMP
   - → DHCP
7. Select Individual packets.
8. Observe the packet list, packet details and packet bytes panes.
9. Analyse the protocol fields.
10. Save the Captured packets.

Result: Thus wireshark was successfully used to capture filter and analyse network packets and their protcol fields.

11/9/26

Ex:5 Wireshark

5. Wireshark Packet Capture



fields.

7/9/24
Ex-6

HAMMING CODE

AIM: To implement error detection and correction using hamming code and verify correction capability using an test data stream.

PROCEDURE:

Sender:

1. Create a sender program.
2. Read a text message from the input field.
3. Convert the text into binary.
4. Calculate & insert the required redundant parity bits.
5. Generate the hamming code.
6. Store the encoded data in a file named channel.

Reciever:

7. Read the encoded data from channel.
8. Recalculate the parity bits.
9. Determine whether an error is present.
10. If an error is detected find its position.
11. Correct the erroneous bit.
12. Remove the redundant bits.
13. Convert the corrected binary data back into ASCII/text.
14. Display the recovered message.

Result: Thus the hamming code was successfully implemented to detect and correct a single-bit error in the transmitted data.

6. Hamming Code (Error Detection & Corr

IDLE Shell 3.10.6

File Edit Shell Debug Options Window Hel

```
Enter 4-bit data : 1011
Hamming Code : 0110011
Received Code : 0110001
Error detected at position : 3
Corrected Code : 0110011
Corrected Data : 1011
>>>
```

the transmitted data.

4/9/26
Ex: 7

Sliding Window

AIM: To implement flow control of the data Link Layer using Sliding window protocol & simulate the transfer of frames between 2 nodes.

Procedure:

Sender:

1. Get the window size from the user.
2. Read a text message.
3. Consider one character as one frame.
4. Assign a frame number to each character.
5. Create frames in the format. [Frame no, Data]
6. Store the frames in Sender buffer.
7. Send and wait for the acknowledgement.
8. Read receiver buffer & check ACK number.
9. If ACK is correct, transmit next frames.
10. If NACK is received, retransmit.
11. Introduce an error by manually modifying a frame/Ack number and observe the behaviour.

Receiver:

1. Read frames from sender buffer.
2. Check frame no., if correct generate ACK.
3. Incorrect, generate NACK
4. Store the result.

Result: Thus the sliding window protocol was successfully implemented & frame flow control using ACK/NACK was verified.

7/9/26
Ex: 7

7. Sliding Window Protocol (Flow Contro

```
Enter window size : 3
Total number of frames : 7

Sending frames...
Frame 0 sent        ACK received
Frame 1 sent        ACK received
Frame 2 sent        ACK lost  ->  Retransmitt
Frame 2 sent        ACK received
Frame 3 sent        ACK received
Frame 4 sent        ACK received
Frame 5 sent        ACK received
Frame 6 sent        ACK received

Transmission completed successfully.
```

& frame flow control using ACK/NACK was verified.

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NMAP Live host Discovery

AIM: To discover live hosts using NMAP scans based on ARP, ICMP, TCP and UDP on the TryHackMe Nmap platform.

Procedure:

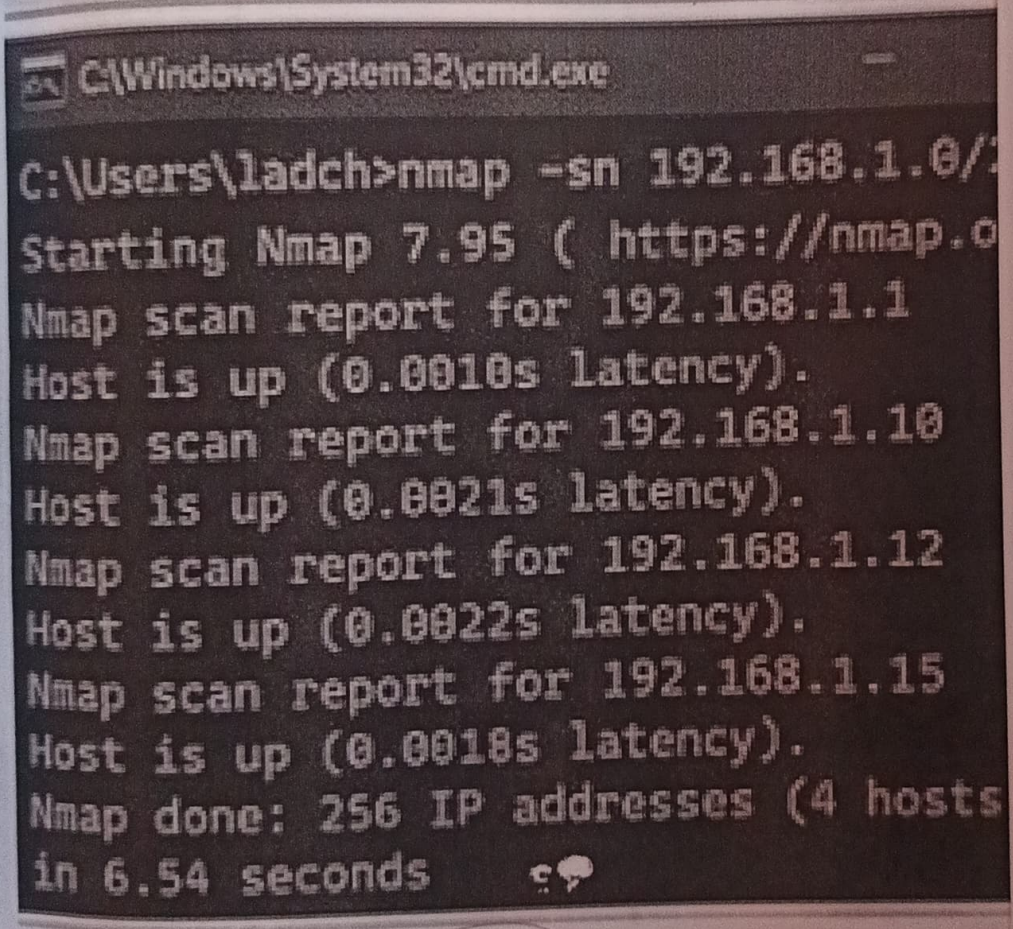
1. Open Nmap room on TryHackMe.
2. Study the target network and Subnet Structure.
3. Identify the target IP address.
4. Perform ARP-based host delivery.
5. Use, nmap - PR - sn target
6. Perform ~~the~~ ICMP echo discovery using, nmap - ICMP timestamp discovery ~~using~~
7. Perform ICMP timestamp discovery using nmap - PP - sn target
8. Perform tcp syn ping using nmap - PS23 Target
9. Perform TCP Ack using the appropriate -PA option.
10. Study UDP-based host discovery.
11. Observe the live hosts identified by Nmap
12. Study reverse DNS lookup using -R.

Result: Thus, Nmap was successfully used to discover live hosts using ARP, ICMP, TCP & UDP based techniques.

15/9/26

1. host Discovery

8. Nmap (Live Host Discovery)



23/9/24

Ex:7 Subnetting using Cisco packet tracer.

Aim: To Implement Classless IP subnetting using cisco packet tracer and establish communication between multiple subnets through routers.

Procedure:

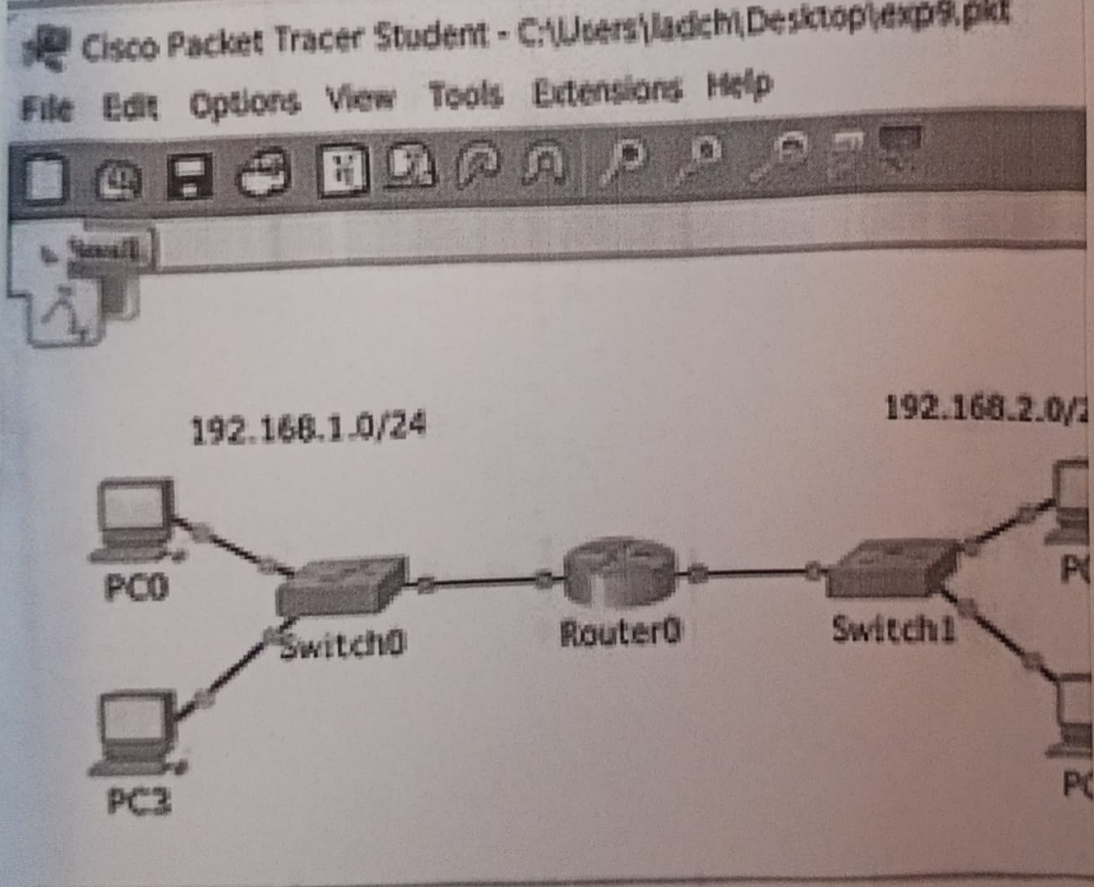
1. Open cisco packet tracer.
2. Create a new network topology.
3. Connect the devices using appropriate Cables.
4. Divide network 192.168.1.0/24 into smaller subnetworks.
5. Use /24 subnet masks as specified in the manual.
6. Configure the router, interface with suitable IP addresses.
7. Configure the switches.
8. Assign IP address and subnet masks and configure the switch ports.
9. Enable router interface using no Shutdown.
10. Test connectivity using ping.
11. Observe packet transmission between subnets.

Result: Thus, classless IP subnetting was successfully implemented in cisco packet tracer & communication between subnet is verified.

22/9/24

Ex.9

9. Subnetting in Packet Tracer



Trace & communication between subnet is verified.

22/9/26

Ex. 10 Internetworking with Routers

AIM: To design and configure an Internetworking using routers in CISCO packet tracer and verify communication between different networks.

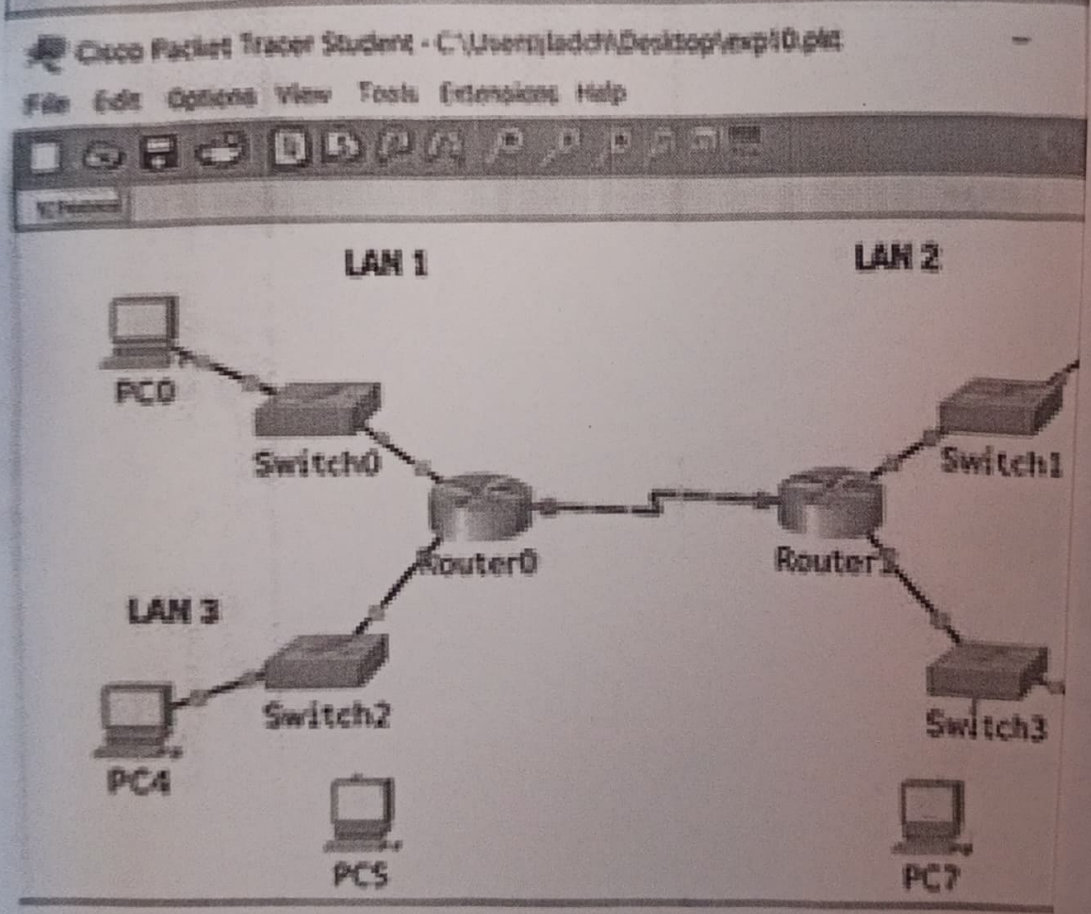
Procedure:

1. Open CISCO packet tracer.
2. Place one router and 2 PCs.
3. Connect the PCs to the router using copper straight through cables.
4. Configure router.
5. Enter privileged mode
   enable.
6. Enter configuration mode
   config t
7. Configure Fast Ethernet 0/0
   Interface fastethernet 0/0
   ip address 192.168.10.1.255.255.255.0
   No shutdown.
8. Configure fast ethernet 0/1:
   Interface fast ethernet 0/1
   ip address 192.168.20.1.255.255.255.0
   no shutdown.
9. Configure PC0:
   Ip address: 192.168.10.2
   Subnet Mask: 255.255.255.0
   Gateway: 192.168.10.1
10. Configure PC1
    Ip address: 192.168.20.2
    Subnet mask: 255.255.255.0
    Gateway: 192.168.20.1
11. Send a PDU from PC0 to PC1
12. Observe the packet transmission and acknowledgement

Result: Thus an Internetwork using a router was successfully designed and configuration & communication between network was verified.

22/9/-

10. Internetworking with Routers



Result: Thus an internetwork using a router was successfully designed and configuration & communication between network was verified

28/9/24

Static Routing and RIP

AIM: To simulate static routing and RIP (Routing Information protocol) using Packet Tracer.

Procedure:

1. Open cisco packet tracer.
2. Plate 2 routers and
3. Connect the routers and PCs.
4. Assign IP addresses to all interfaces.
5. Configure router interfaces using IP address and no shutdown.
6. Add static router using
   ip route <destination-network> <subnet-mask> <next-hop>
7. Verify the routing table using show ip route.
8. Test connectivity using ping.

RIP:

9. Configure the router interfaces
10. Enter RIP configuration mode:
    router rip
    version 2
11. Add the connected networks
    network <network address>
12. Repeat the configuration on the other router.
13. Verify routing table.
14. Ping a host in the remote network.
15. Thu

Result: Thus, static routing and RIP was successfully configured and the connectivity between different network was notified.

EX. NO: 11 28/9/26

11. Routing at Network Layer (Static &

Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
Router#configure terminal
Router(config)#ip route 192.168.2.0 255.255.255.0 1
Router(config)#router rip
Router(config-router)#network 192.168.1.0
Router(config-router)#network 192.168.2.0
Router#show ip route
Codes: C - connected, S - static, R - RIP

C     192.168.1.0/24 is directly connected, Gig0/
C     10.0.0.0/30 is directly connected, Gig0/1
S     192.168.2.0/24 [1/0] via 10.0.0.2
R     192.168.3.0/24 [120/1] via 10.0.0.2
Router#
```

lee

TCP/UDP Socket Communication

AIM: To implement end-to-end communication the transport layer using TCP/UDP sockets and develop a simple chat application.

Procedure:

Echo client-server

1. Create a server program
2. Create a client program and a socket at the server.
3. Bind socket to an Ip address and post.
4. Make the server listen for client connections.
5. Accept client connections.
6. Client sends a message.
7. Server recieves and echos the message.
8. Display the response at the client.

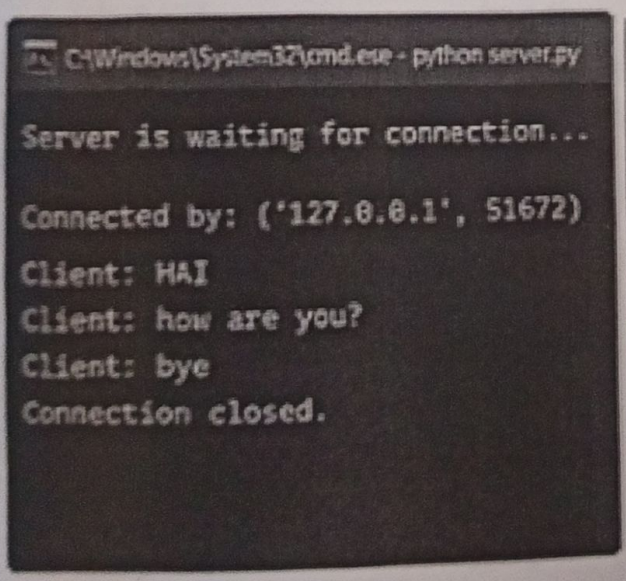
Chat program:

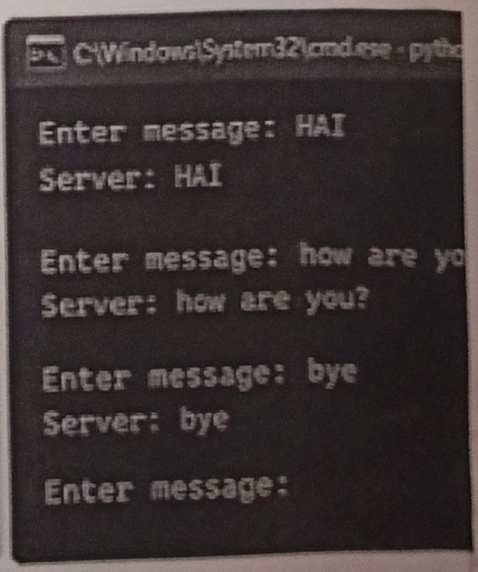
9. Create Client & server socket programs.
10. Establish a connection.
11. Read messages from the user.
12. Send messages through the socket.
13. Recieve and display messages at the other end.
14. Continue communication until connection is terminated.

Result: Thus, end-to-end communication was successfully implemented using socket programming & echo/chat communication between client and server was verified.

Ex: 12 TCP/UDP

12. End–End Communication (TCP/UDP Sockets)





and server was written.

5/10/24
13

OWN PING Program

AIM: To Implement a simple ping program using socket programming and verify the reachability of a destination host.

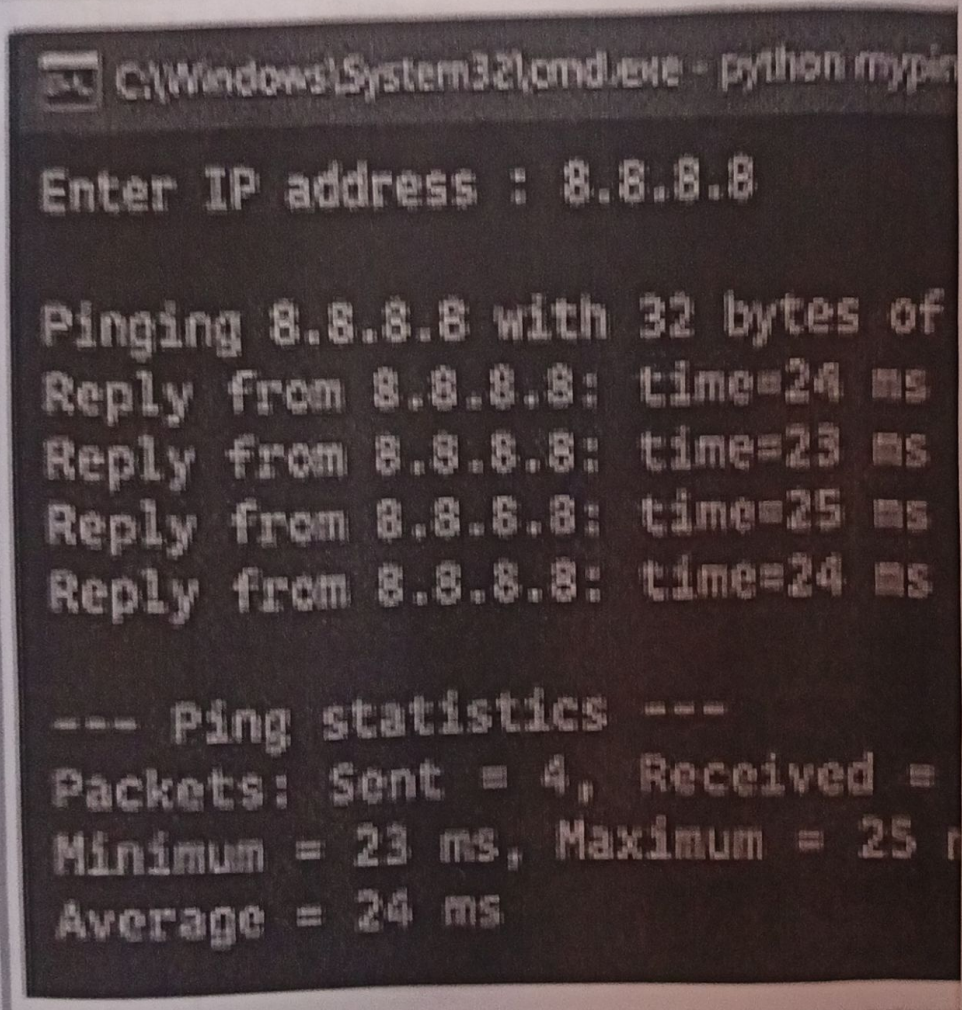
PROCEDURE:

1. Create a program to accept a destination IP address.
2. Generate a socket for network communication
3. Generate an ICMP echo request.
4. Send the request to the destination
5. Wait for the ICMP echo reply.
6. Calculate the responsible time.
7. Display whether the host is reachable.
8. Repeat the process for multiple requests.
9. Display the process transmission statistics.

Result: Thus a simple ping program was successfully implemented and the reachability of the destination host was verified.

5/10/26

13. Own Ping Program



successfully Implemented and the reachability of the destination host was confirmed.

5/10/26

14 RAW SOCKET PACKET SNIFFER

AIM: To Implement packet sniffer using raw sockets and analyze the packets received through the network interface.

PROCEDURE:

1. Create a raw socket using the socket programming interface.
2. Select the required network interface.
3. Enable reception through the raw socket.
4. Capture incoming network packets.
5. Read the packet header.
6. Extract important fields such as:
   - → Source IP
   - → Destination IP
   - → Protocol
   - → Packet Length
7. Identify the protocol captured packet.
8. Display packet information.
9. Continue capturing packets until the program is terminated.

completed & 13/10/26

Result: Thus, RAW sockets were successfully used to capture and analyze network

14. RAW Socket Packet Sniffing

